

DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

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CLASS

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INDEX
NUMBER

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PRELIMINARY EXAMINATION 2023 SECONDARY 4 EXPRESS

MATHEMATICS

4052/01

Paper 1

22 August 2023

2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Calculator Model

Answer **all** questions.

If working is needed for any question it must be shown with the answer.

Omission of essential working will result in loss of marks.

The total of the marks for this paper is 90.

The use of an approved scientific calculator is expected, where appropriate.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π , use either your calculator value or 3.142, unless the question requires the answer in terms of π .

Total Marks

Mathematical Formulae*Compound interest*

$$\text{Total amount} = P \left(1 + \frac{r}{100} \right)^n$$

Mensuration

$$\text{Curved surface area of a cone} = \pi r l$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3} \pi r^3$$

$$\text{Area of a triangle } ABC = \frac{1}{2} ab \sin C$$

$$\text{Arc length} = r\theta, \text{ where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$

Trigonometry

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Statistics

$$\text{Mean} = \frac{\Sigma fx}{\Sigma f}$$

$$\text{Standard deviation} = \sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f} \right)^2}$$

Answer **all** the questions.

- 1 Given that $5^x = \sqrt[3]{25}$, find x .

Answer $x = \dots\dots\dots$ [1]

- 2 Solve $kx^2 + (2k + 3)x + 6 = 0$ by factorisation.

Answer $x = \dots\dots\dots$ or $\dots\dots\dots$ [2]

- 3 It is given that $(3.4 \times 10^m) \div (5.3 \times 10^n) = a \times 10^p$, where $1 \leq a < 10$.
Find p in terms of m and n .

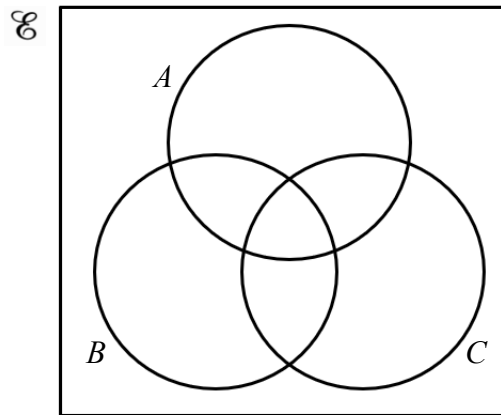
Answer $p = \dots\dots\dots$ [2]

- 4 (a) Describe the elements of the set $\{1, 4, 6, 8, 9\}$.

.....

..... [1]

- (b) On the Venn diagram, shade the region which represents $(A \cup B) \cap C'$.



[1]

- 5 The magnetic force, F , of a magnet is inversely proportional to the square of the distance, d , from the magnet.
The distance of the magnet is decreased by 60%.

Calculate the percentage increase in the magnetic force.

Answer % [2]

- 6 The area of a triangle ABC is 14 cm^2 .
 $AB = 7 \text{ cm}$ and $BC = 5 \text{ cm}$.

Find the largest possible value for the angle ABC in degrees.

Answer ° [2]

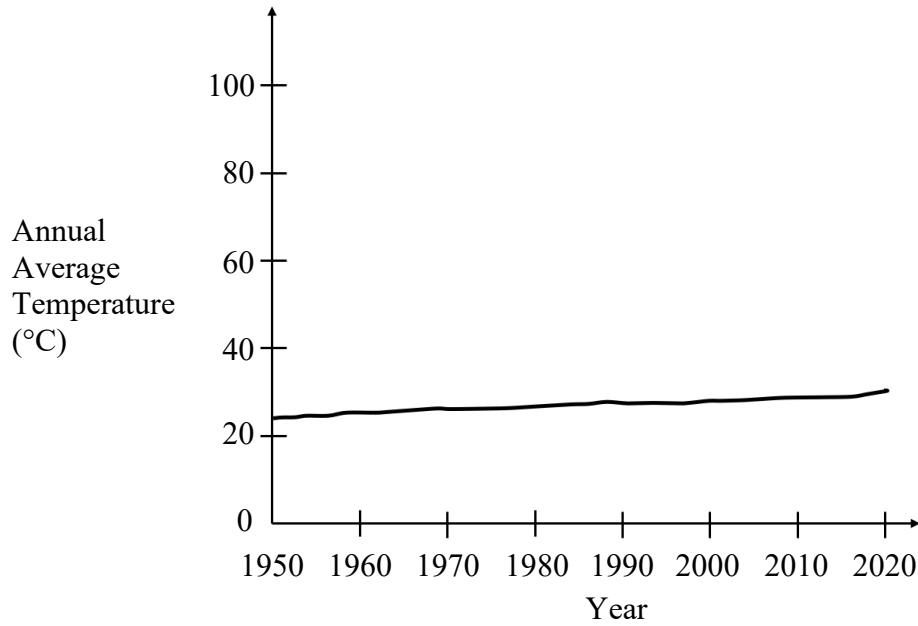
- 7 Siva writes the numbers
3, 5, 7, 11, x , y , z .

The mode is 11, the median is 8 and the range is 10.

Find the value of $x + y + z$.

Answer [3]

- 8 The graph shows the average annual temperature in Singapore over the last 70 years.



- (a) State one misleading feature of the graph.

..... [1]

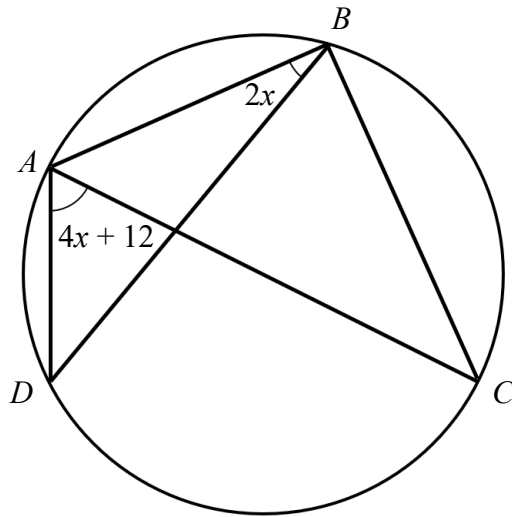
- (b) Explain how this feature affects the reader's interpretation of the graph.

..... [1]

- 9 Write as a single fraction in its simplest form $\frac{3}{2-x} + \frac{x}{x^2-3x+2}$.

Answer [3]

10



The diagram shows a circle $ABCD$.

AC is a diameter of the circle, angle $CAD = (4x + 12)^\circ$ and angle $ABD = 2x^\circ$.

(a) Find, in terms of x , angle CBD .

Answer [1]

(b) Find x .

Answer $x =$ [2]

(c) Mark a point Q on the diagram such that angle $BQD =$ angle BAD . [1]

- 11 (a)** Express $6x - 11 + x^2$ in the form $(x + p)^2 + q$.

Answer [2]

- (b)** Hence solve $6x - 11 + x^2 = 0$.

Answer $x =$ or [2]

- 12** In Singapore, David pays \$2.88 for one litre of milk.
On a visit to Los Angeles, he paid 7.90 US dollars for one gallon of milk.

1 US dollar = 1.34 Singapore dollars.

1 gallon = 3.785 litres.

Is milk cheaper in Singapore or Los Angeles?

You must show your calculations.

Answer

[3]

- 13** The conversion of temperature from degrees Celsius to degrees Fahrenheit can be done with this formula.

$$F = \frac{9}{5}C + 32 ,$$

where F is the temperature in degrees Fahrenheit
and C is the temperature in degrees Celsius.

- (a) The difference in temperature between two cities is 27 degrees Fahrenheit.

Find the difference in temperature between these two cities in degrees Celsius.

Answer degrees [1]

- (b) Adin claims that the temperatures in degrees Fahrenheit and degrees Celsius are directly proportional.

Is he correct? Justify your answer.

.....

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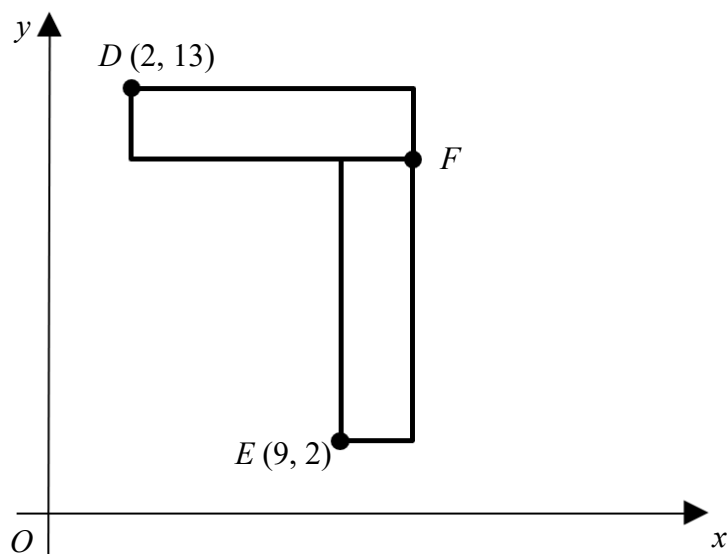
..... [1]

- 14** Ali invested \$10 000 in an account paying compound interest at r % per year.
After 7 years, his investment had earned a total interest of \$3159.31.

Find the value of r .

Answer $r =$ [3]

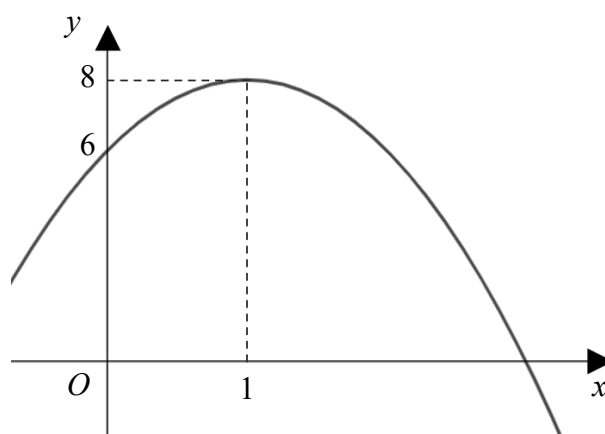
- 15 The diagram shows two congruent rectangles.
The sides are horizontal and vertical.
Point D has coordinates $(2, 13)$ and E has coordinates $(9, 2)$.



Find the coordinates of F .

Answer (.....,) [2]

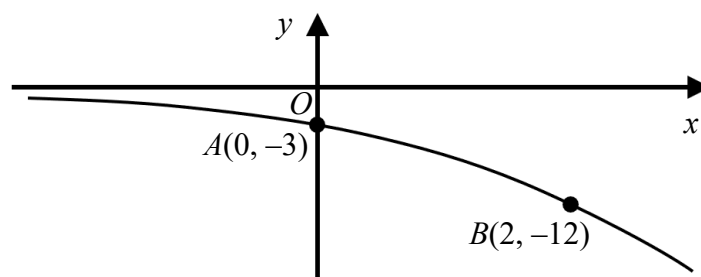
- 16 The diagram shows a quadratic graph.
The turning point has coordinates (1, 8) and the y -intercept has coordinates (0, 6).



Find the equation of the quadratic graph in the form $y = ax^2 + bx + c$.

Answer $y = \dots\dots\dots$ [3]

- 17 The sketch shows the graph of $y = -ka^x$.
The points $A(0, -3)$ and $B(2, -12)$ lie on the graph.



- (a) Find the value of k and of a .

Answer $k = \dots\dots\dots$

$a = \dots\dots\dots$ [2]

- (b) Find the length of the straight line drawn from A to B .

Answer $\dots\dots\dots$ units [2]

- 18 (a) Express 252 as a product of its prime factors.

Answer [1]

- (b) Given that k is a prime number, find the value of k such that $252k$ is a perfect square.

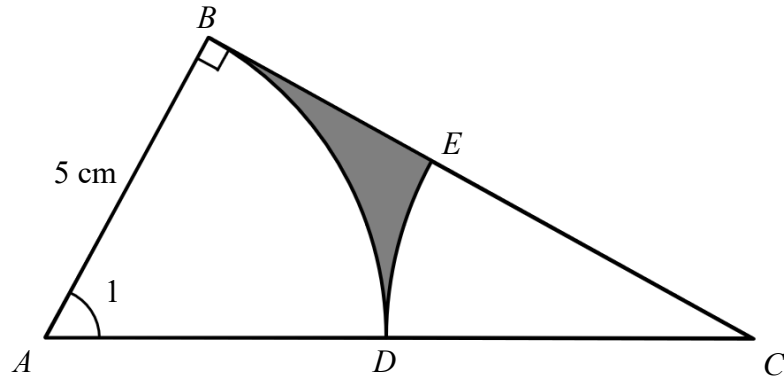
Answer $k =$ [1]

- (c) The highest common factor (HCF) of 252 and the number X is 2×3^2 .
The lowest common multiple (LCM) of 252 and the number X is $2^2 \times 3^3 \times 5 \times 7$.

Find the value of the number X .

Answer [1]

- 19 The diagram shows a right-angled triangle ABC with angle $ABC = 90^\circ$.
 D is a point on AC and E is a point on BC .
 ABD is a sector of the circle with centre A and CED is a sector of the circle with centre C .
 $AB = 5$ cm and angle $BAC = 1$ radian.



- (a) Find the length of CD .

Answer cm [2]

- (b) Find the perimeter of the shaded region.

Answer cm [4]

20 A map of Southeast Asia has a scale of 1 : 5 000 000.

(a) The length of a river on the map is 18.6 cm.

Calculate the actual length, in kilometres, of this river.

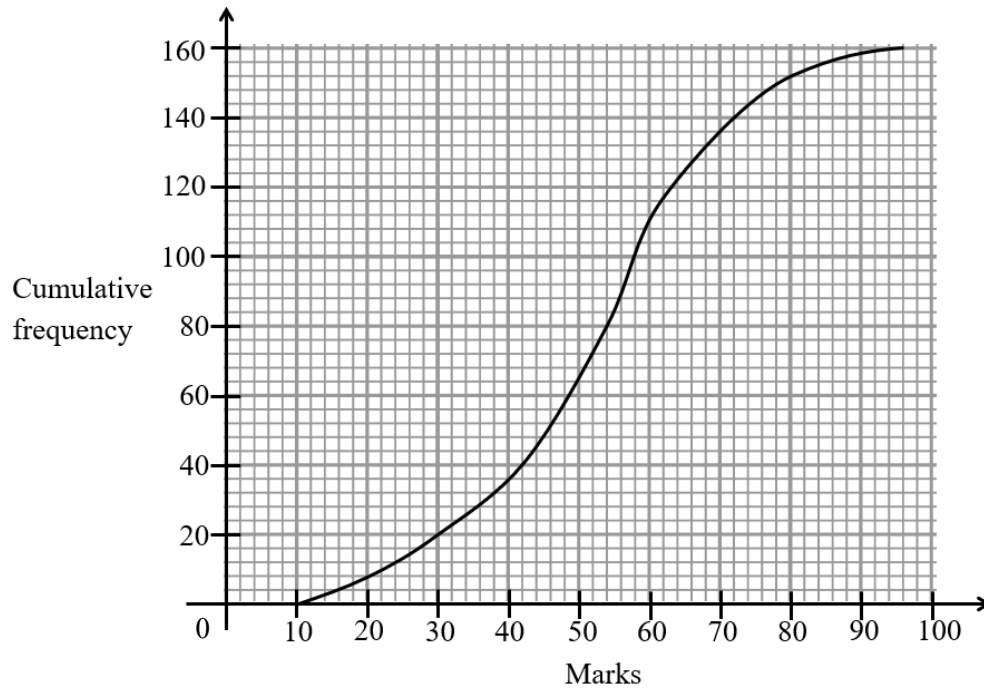
Answer km [1]

(b) The area of Brunei is 5765 km².

Calculate the area, in square centimetres, of Brunei on the map.

Answer cm² [2]

- 21** The cumulative frequency curve shows the distribution of the marks of 160 students in a Mathematics examination.



Use the curve to

- (a)** state the highest mark,

Answer [1]

- (b)** estimate the 30th percentile,

Answer [1]

- (c)** find the probability that a student, chosen at random, has at least 70 marks.

Answer [2]

22

$$a = \frac{kr^2}{r^2 - x}$$

Rearrange the formula to make r the subject.

Answer $r = \dots\dots\dots$ [3]

23 A is the point $(3, -2)$ and $\overrightarrow{AB} = \begin{pmatrix} 7 \\ -3 \end{pmatrix}$.

(a) Find $|\overrightarrow{AB}|$.

Answer units [1]

(b) Find the equation of the line AB .

Answer [2]

(c) C is the point $(-3, 14)$ and D is the point $(11, 8)$.

Use vectors to show that AB is parallel to CD . Justify your answer.

Answer

[3]

- 24 (a) Jane runs a community service.
 17 of the members are paid staff and 23 are volunteers.
 Her aim is that at least 70% of the members should be volunteers.

Find the smallest number of volunteers that she would need to recruit in order to achieve her aim.

Answer [3]

- (b) A shop sells eggs at 20% profit of the cost price.
 To clear stocks, the owner decides to put a 20% discount on the selling price of the eggs.

Does the shop make a loss? Justify with mathematical reasoning.

.....

 [2]

- 25** Different rooms in a hotel cost different prices to rent on weekdays and weekends.
 On a weekday, a single room costs \$80, a double room cost \$130 and a deluxe room cost \$170.
 On a weekend, a single room costs \$120, a double room cost \$180 and a deluxe room cost \$230.

(a) Represent this information in a 2×3 matrix $\mathbf{P}$.

Answer $\mathbf{P} =$ [1]

(b) The hotel has a total of 60 single rooms, 40 double rooms and 15 deluxe rooms.

Evaluate the matrix $\mathbf{Q} = \mathbf{P} \begin{pmatrix} 60 \\ 40 \\ 15 \end{pmatrix}$.

Answer $\mathbf{Q} =$ [2]

(c) Explain what each element in $\mathbf{Q}$ represents.

.....

..... [1]

(d) The elements of the matrix $\mathbf{R}$, where $\mathbf{R} = \mathbf{SP}$, represents the total costs for renting each of the different rooms for 3 weekdays.

Write down the matrix $\mathbf{S}$.

Answer $\mathbf{S} =$ [1]

- 26** A bag contains 7 red marbles, 5 blue marbles and 2 yellow marbles.
When a red marble is chosen, it is replaced.
When a blue marble is chosen, it is not replaced.
When a yellow marble is chosen, it is not replaced and a red marble is added to the bag.
Two marbles are chosen at random.

Find, as a fraction in its simplest form, the probability that

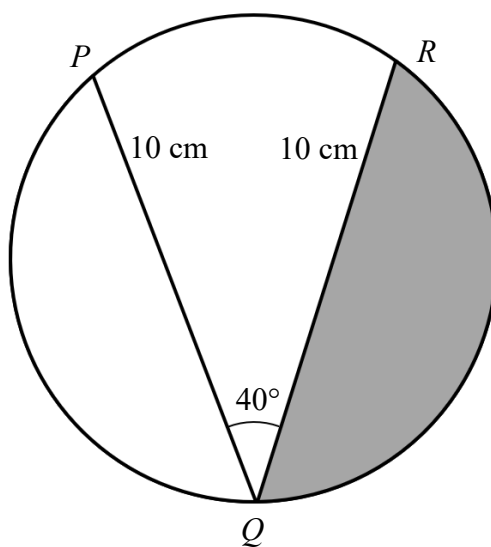
- (a)** two blue marbles are chosen,

Answer [2]

- (b)** the second chosen marble is yellow.

Answer [3]

- 27 PQR are points on a circle.
 $PQ = QR = 10$ cm and angle $PQR = 40^\circ$.



Find the area of the shaded region.

Answer cm^2 [6]

END OF PAPER